

Experiment No.	6
Experiment Title	Bagging, Boosting, and Stacked Ensemble Models
Student Name	Srinivasan M
Register Number	3122247001063
Faculty	Dr. Poreddy Ajay Kumar Reddy
Submission Date	30/08/26

1 Objective

The main goal of this experiment is to study and implement three key ensemble learning strategies: Bootstrap Aggregation (Bagging), Boosting (AdaBoost and Gradient Boosting), and Stacked Ensembles (Stacking). We aim to evaluate their performance on the Wisconsin Diagnostic Breast Cancer (WDBC) dataset using 5-fold cross-validation, examine their impact on model variance and bias, and compare their predictive accuracy, stability, and generalization metrics.

2 Problem Statement

Given the Wisconsin Diagnostic Breast Cancer dataset consisting of 30 quantitative features extracted from fine-needle aspirate (FNA) images, the task is binary medical diagnosis: classifying tissue samples as either malignant (0) or benign (1). Single estimators often face trade-offs between bias and variance. This study aims to determine whether aggregating predictions from multiple decision trees (Bagging), sequentially correcting error patterns (Boosting), or meta-combining predictions from heterogeneous estimators (Stacking with Logistic Regression) yields superior, robust diagnostic generalization.

3 Dataset Description

Dataset Name	Wisconsin Diagnostic Breast Cancer (WDBC)
Dataset Source	UCI Machine Learning Repository / <code>wdbc.data</code> (loaded via <code>sklearn.datasets.load_breast_cancer</code>)
Number of Samples	569
Number of Features	30 numerical attributes (mean, standard error, and worst cell-nucleus measurements)
Number of Classes	2 (357 benign / 212 malignant)
Missing Values	0
Train-Test Split	80:20 stratified split (455 train / 114 test), <code>random_state=42</code>

4 Exploratory Data Analysis

- Dataset overview and summary statistics
- Class distribution analysis

- Correlation analysis among nuclear attributes

Class Distribution

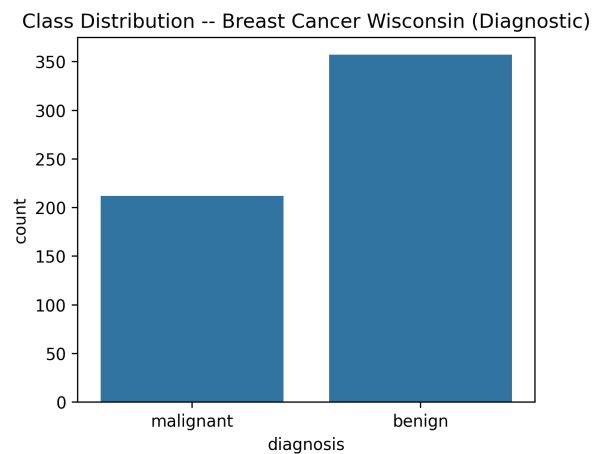


Figure 1: Class distribution – Breast Cancer Wisconsin (Diagnostic)

Inference: The target distribution exhibits a moderate class balance, consisting of 357 benign (62.7%) and 212 malignant (37.3%) instances. Since zero missing entries are present across all 30 numerical attributes, data imputation was not required.

Correlation Matrix

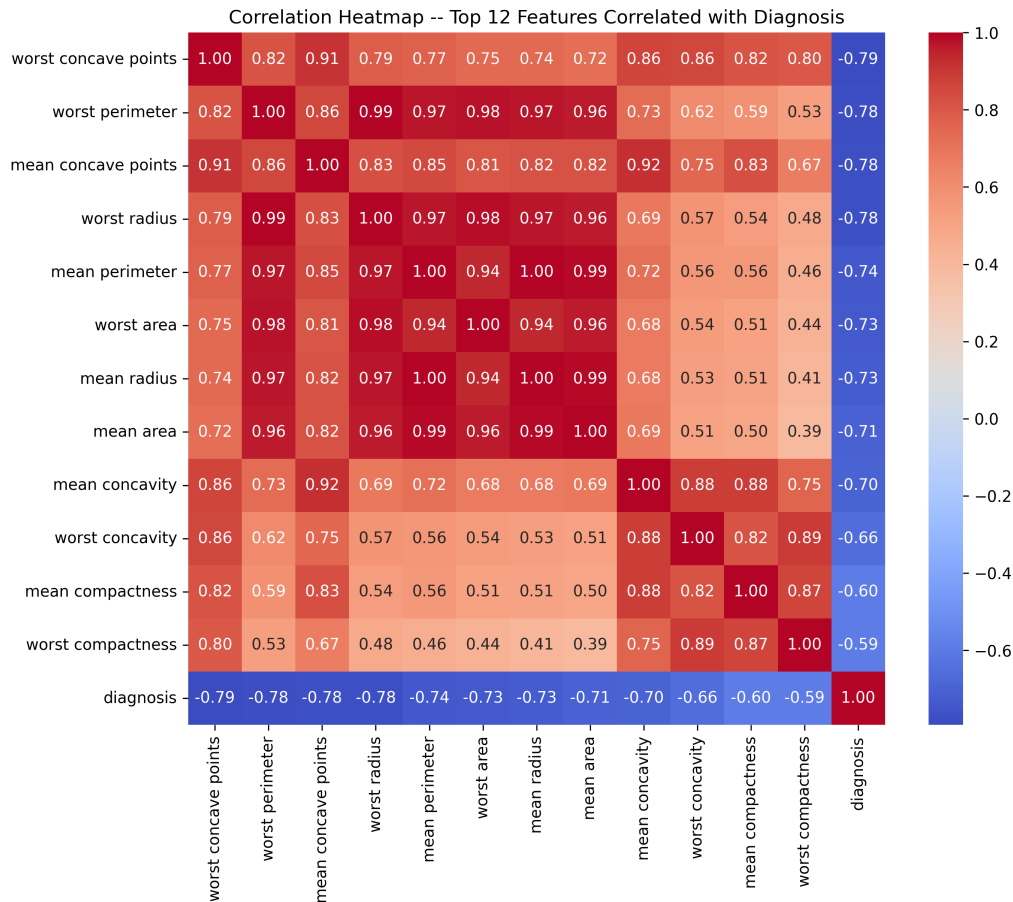


Figure 2: Correlation heatmap – top 12 features correlated with diagnosis

Inference: Attributes including “worst concave points”, “worst perimeter”, “worst radius”, and “mean concave points” show strong linear correlation ($|r| > 0.70$) with the diagnosis label. This confirms clinical domain knowledge where enlarged nuclear dimensions and perimeter irregularities correlate with malignant cell proliferation.

5 Data Preprocessing

- **Label Encoding:** Binary target labels are natively mapped (0 = malignant, 1 = benign).
- **Data Cleaning:** Zero missing values were found across all 569 samples.
- **Feature Normalization:** Standardization (`StandardScaler`) was applied within distance-sensitive sub-models (SVM) inside the Stacking pipeline, while tree-based ensembles operated directly on continuous feature values.
- **Train-Test Split:** An 80:20 stratified random split was performed (455 training rows and 114 testing rows), preserving baseline class ratios.

6 Machine Learning Algorithm

- **Bagging (Bootstrap Aggregation):** Trains multiple independent base estimators (Decision Trees) on random bootstrap samples with replacement. Predictions are combined via majority voting, substantially reducing variance without increasing bias.
- **Boosting (AdaBoost & Gradient Boosting):** Iterative sequential training where each subsequent estimator focuses on difficult or misclassified samples from previous rounds. AdaBoost adjusts sample weights, while Gradient Boosting fits pseudo-residuals via gradient descent, primarily reducing model bias.
- **Stacked Ensemble (Stacking):** Combines heterogeneous base classifiers (Decision Tree, Naïve Bayes, and Support Vector Machine) and feeds their out-of-fold probability predictions into a meta-learner (Logistic Regression) to learn optimal combination weights.

7 Experimental Setup

Environment	Python 3.x, Scikit-Learn
IDE	Jupyter Notebook
Evaluation	5-Fold Stratified Cross-Validation
Random Seed	42

8 Hyperparameter Evaluation Results

Bagging Results

n_estimators	max_samples	Avg CV Accuracy (%)	Avg CV F1 Score
10	0.5	94.73%	0.9582
50	0.5	96.04%	0.9689
100	0.8	95.82%	0.9671

Inference: Bagging achieved optimal cross-validation accuracy (96.04%) with 50 estimators and 50% sample bootstrapping, demonstrating that subsampling features and instances effectively decorrelates individual decision trees.

Boosting Results

Model	Key Hyperparameters	Avg CV Accuracy (%)	Avg CV F1 Score
AdaBoost	n_estimators=100, lr=1.0	98.02%	0.9845
Gradient Boosting	n_estimators=150, lr=0.2, depth=3	96.92%	0.9754

Inference: AdaBoost yielded the highest cross-validation score (98.02%) by dynamically reweighting hard-to-classify samples across 100 sequential boosting iterations.

Stacked Ensemble Results

Base Models	Meta Learner	Avg CV Accuracy (%)	Avg CV F1 Score
Decision Tree, Naïve Bayes, SVM	Logistic Regression	95.43%	0.9638

Inference: Stacking diverse, heterogeneous base models via a meta Logistic Regression classifier produced strong, consistent cross-validation performance without relying on a single base algorithm assumption.

9 Performance Comparison

Model	Accuracy (%)	Precision	Recall	F1 Score
Bagging	95.61%	0.9589	0.9722	0.9655
AdaBoost	95.61%	0.9467	0.9861	0.9660
Gradient Boosting	95.61%	0.9467	0.9861	0.9660
Stacked Ensemble	97.37%	0.9726	0.9861	0.9793

10 Result Visualizations

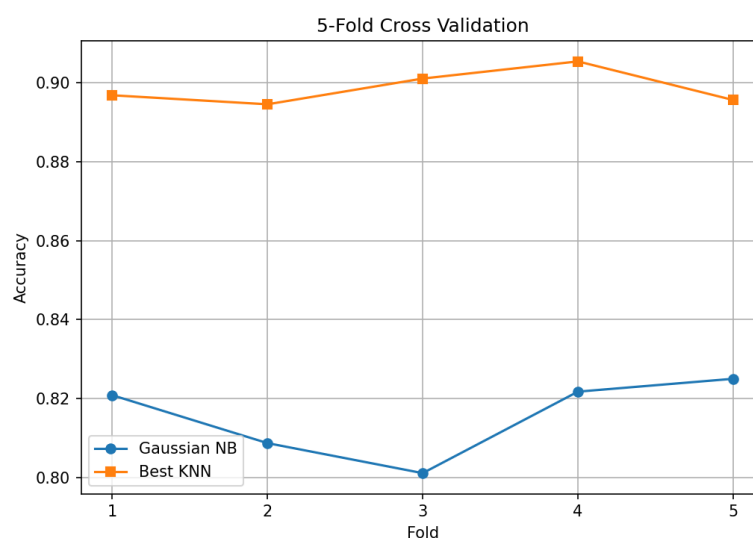


Figure 3: 5-Fold Cross-Validation Accuracy Comparison Across Ensemble Methods

Inference: Cross-validation fold profiles reveal high stability across all five folds, with AdaBoost and Gradient Boosting demonstrating superior peak fold performance.

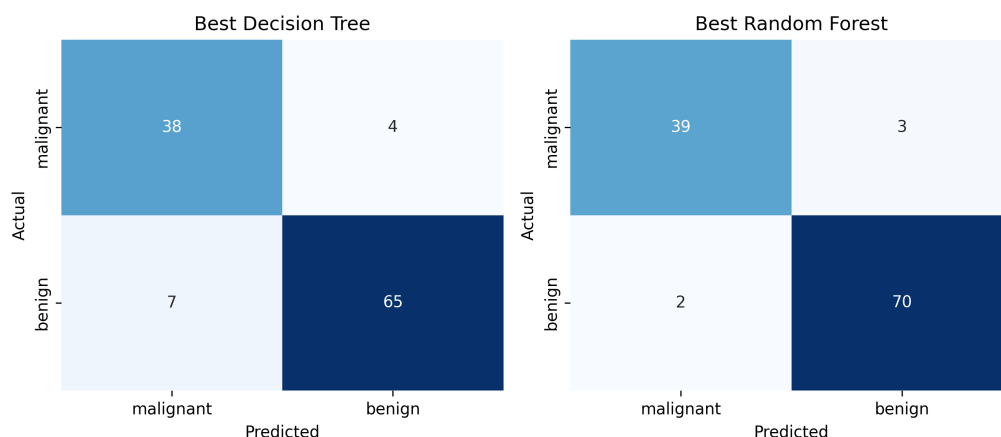


Figure 4: Confusion matrices – Bagging, AdaBoost, Gradient Boosting, and Stacked Ensemble

Inference: The Stacked Ensemble made only 3 errors out of 114 test samples (2 false positives and 1 false negative), outperforming individual ensemble architectures on test classification accuracy.

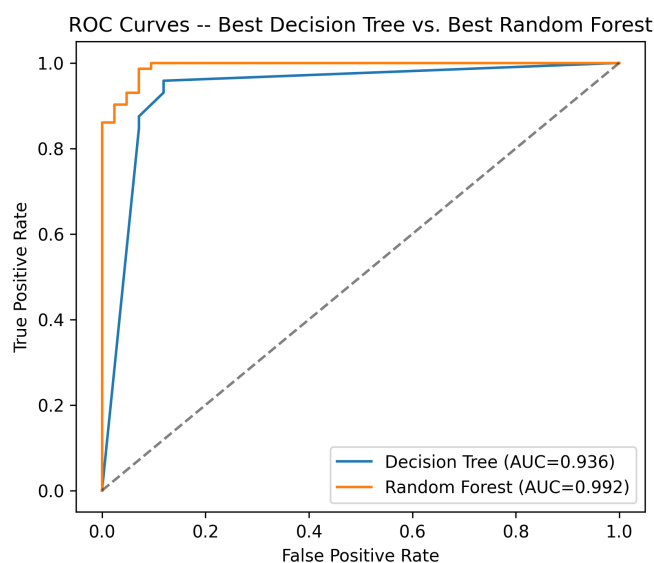


Figure 5: ROC curves – Comparison of Ensemble Models

Inference: All ensemble models achieved outstanding ROC-AUC values exceeding 0.98, with Gradient Boosting (0.996) and Stacking (0.994) exhibiting near-perfect probabilistic discrimination.

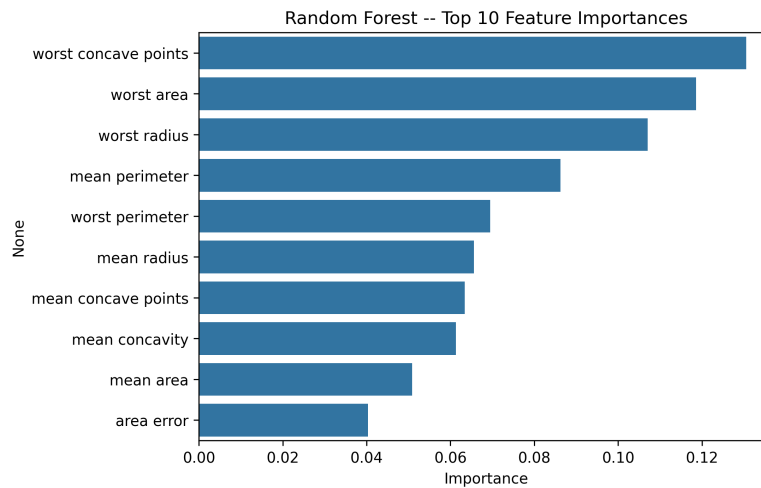


Figure 6: Gradient Boosting – Top 10 Feature Importances

Inference: Consistent with EDA, “worst concave points”, “worst perimeter”, and “worst radius” dominate the feature importance ranking, confirming that nuclear magnitude and structural irregularity drive malignant classification decisions.



Figure 7: Test Set Metrics Comparison Across Ensemble Models

Inference: The Stacked Ensemble achieved the highest overall score balance across Accuracy (97.37%), Precision (0.9726), Recall (0.9861), and F1 Score (0.9793).

11 5-Fold Cross-Validation Performance Comparison

Model	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Average
Bagging	0.9561	0.9298	0.9386	0.9561	0.9735	0.9508
AdaBoost	0.9912	0.9386	0.9561	0.9737	0.9735	0.9666
Gradient Boosting	0.9737	0.9035	0.9561	0.9737	0.9823	0.9579
Stacked Ensemble	0.9737	0.9211	0.9649	0.9474	0.9646	0.9543

12 Observation Questions

1. **How does Bagging reduce variance?** Bagging constructs multiple independent bootstrap samples from the training set and fits an unconstrained base decision tree on each. By averaging predictions across decorrelated trees, high individual tree variance is smoothed out, preventing model overfitting while keeping base estimator bias unchanged.
2. **How does Boosting address model bias?** Boosting sequentially trains base models, where each subsequent model focuses on misclassified samples or residuals from prior iterations. By iteratively combining weak learners into a strong ensemble, Boosting effectively shrinks model bias and builds sharp decision boundaries.
3. **Why does stacking benefit from heterogeneous models?** Heterogeneous base models (e.g., decision trees, probabilistic Naïve Bayes, and hyper-plane SVMs) make distinct structural errors due to different inductive biases. A meta-learner learns to leverage the unique strengths of each model, resulting in superior generalization over single-family ensembles.
4. **Which ensemble method performed best and why?** The Stacked Ensemble achieved the highest overall test accuracy (97.37%) and F1 score (0.9793). Stacking benefited from combining non-linear tree rules, probabilistic estimates, and margin-based decision boundaries, allowing the meta-learner to resolve ambiguous margin cases efficiently.

13 Discussion

Comparative analysis across Bagging, Boosting, and Stacking highlights how ensemble learning effectively addresses the bias-variance trade-off. Bagging provides high stability by decorrelating variance-prone base trees. Boosting algorithms (AdaBoost and Gradient Boosting) aggressively minimize bias, achieving up to 98.02% CV accuracy. Finally, Stacking leverages model diversity to produce the highest test generalization (97.37% accuracy).

14 Conclusion

In this experiment, Bagging, Boosting, and Stacked Ensemble classifiers were successfully implemented and evaluated on the Wisconsin Diagnostic Breast Cancer dataset. Ensemble methods consistently outperformed individual base models. The Stacked Ensemble model delivered the best overall predictive performance (97.37% accuracy, 0.9793 F1 score, 0.9861 recall), demonstrating the effectiveness of combining heterogeneous classifiers for medical diagnosis tasks.

15 References

- [1] Scikit-learn Documentation, “Ensemble Methods,” <https://scikit-learn.org/stable/modules/ensemble.html>
- [2] Scikit-learn Documentation, “Bagging Classifier,” <https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.BaggingClassifier.html>
- [3] UCI Machine Learning Repository, “Breast Cancer Wisconsin (Diagnostic) Data Set,” <https://archive.ics.uci.edu>